



Assignment No:- 1

Title of Experiment:-

To design and configure a Local area Network (LAN), analyze network traffic using Wireshark packet analyzer tool, test connectivity with standard Network commands and demonstrate file transmission between

Instructor Name:-

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DOP:-

POS:-

Objectives.

Demonstrate network types, topology & transmission media using Cisco Packet Tracer.

Equipment and software.

Hardware

- ① Computers/Laptops with Network Interface Card (NIC)
- ② Ethernet cables (RJ-45)
to physically connect computers to switches or routers
- ③ Network Switch.
Connects multiple computers within same LAN.
- ④ Wireless Router.
Provides wireless connectivity & connects Local Network to internet.
- ⑤ Storage - to store packet capture files generated by Wireshark.

Software

- ① Cisco Packet Tracer
used to design, simulate & configure computer network, creates virtual network, used before building network.



- ② Wireshark network protocol analyzer used to capture and inspect network packets in real time
- ③ OS (Windows / Linux)
- ④ Command Prompt / Terminal
- ⑤ File transfer facility.

Theory

- Introduction:- Computer Network is collection of inter connected computers and Networking devices that communicate with each other to exchange data share resources such as files, printers & internet connectivity.

Network devices like Hub, switch, router play important role in transferring data between devices. Physical connection is established using ethernet cable RJ-45, prepared using crimping tool for designing & simulating Networks, Cisco Packet Tracer is used, Wireshark is used to capture & analyze packets transmitted over network.

LAN (Local Area Network)
It is a computer network that connects devices within a limited geographical area like laboratory, office, building or campus.



4 IP Addressing

An IP Address uniquely identify device on network.
For communication within LAN devices must be configured with valid IP address belonging to same network.

Device	IP Address	Gigaset Host
PC0	192.168.0.3	255.255.255.0
PC1	192.168.0.2	255.255.255.0
PC2	192.168.1.1	255.255.255.0

∴ devices can communicate because they belong to same IP network.

MAC address.	7
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A mac (media access control) Address is an physical Address of a me network interface card (NIC). It is permanently assigned by the manufacturer & is used by switches to identify devices within same local network.

◆ Networking defines

Hardware component used to connect computers & other electronic devices to form networks

① Substanz

- Connects devices within same network (usually LAN)
- like computers, printers, servers, etc.

- operates at data link layer of OSI model & sometimes in network layer to perform routing.



① Router

- connects multiple network together, such as connecting LAN to Internet.
- forwards data packets (Networks)



② Hub

- connects multiple computers so it broadcasts data to all connected devices without checking the destination. (works at physical layer)

Transmission Media

Physical or wireless path through which data signals travel from sender to receiver (Computer Network (Physical Layer)).

Types

Guided (Wired) Trans- ← Unguided (Wireless) Transmission Media → Transmission Media

① Guided (Wired) Transmission Media

i) Twisted Pair Cables :-

- ① Unshielded TP (UTP) :- used for Ethernet network. Categories include cat 5, cat 6, cat 7 & etc.
- ② STP :- use shielding to reduce EMI



ii) Coaxial cables

- used in Television & early CN.



iii) optic fibre cables

- Transmit data as light offering high-speed & long-distance communication.



- ① single-mode fibre (SMF) :- for long distance
- ② Multi-mode fibre (MMF) :- for shorter distance

ii) Ethernet cables

- special type of UTP cable used in LAN including Cat 5e, Cat 6, etc, which defines speed, frequency.
- RJ-45 connector :- 8-pin connector attached to both ends of an Ethernet cable to connect to switches, routers, or computer

iii) Crimping tool

Primary purpose is to connect RJ45 connectors to the end of ethernet cables (cat 5e, cat 6)

iv

Network Topology. Physical or logical arrangement of devices in a network.

①

Bus Topology.

- All devices connected to single communication cable.
- Data transmitted by one device travels along the shared medium.



②

Star Topology.

All devices are connected to central device like switch or hub. Central devices manages communication between connected device, in our experiment we use star topology to design & configure LAN.



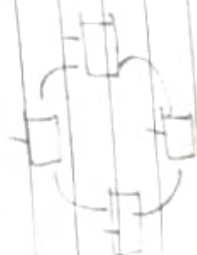


Adv:- failure of one connecting device/cable only affect corresponding device.

Disadv:- failure of central switch can affect the entire network.

③ Ring Topology

- Each device connected to other device forming closed loop.
- Data travels from one device to another around ring.



④ Mesh Topology

- Devices are connected to multiple or all other devices.
- It provides multiple path for data transmission.



⑤ Tree Topology

- Hierarchical arrangements of multiple star networks.
- Suitable for large networks with diff level of organisation.



Procedure

[Step 1] Create network topology.

- open Cisco Packet Tracer. Place one cisco 2950-24 switch & 4 devices (2 PC, 2 laptop).
- verify copper-straight-through cables connect all devices forming star topology.

[Step 2] configure IP addresses.

- open each device (desktop → IP config)

- Assign static IP addresses to all devices within subnet.

[Step 3] Verify connectivity.

- open command prompt on PC.

- execute ping to test connectivity with other devices.

[Step 4] Analyze packet transmission.

- switch simulation mode

- Add PCV to send packet. observe packet through switch.

[Step 5] observe OSI layer information.

- open PCV information

- view OSI Model tab to observe packet processing &

verify source IP, destination IP.

observations & Result

Wireshark Packet Analyzer Tool.

- open source packet analyzer (it allows to capture & instantaneously browse the traffic occurring on a comp. network in real time)

launch Wireshark → look for traffic/graph

click stop capture ← terminal ← start capture
(red square icon) (ping -) (green shake fit icon)

Apply Filter → Analyze Protocol Stack
(Packet details)

ex. Filter syntax	example.
1. $ip.addr == x.x.x.x$	filter by host $ip.addr == 192.168.0.1$
2. $tcp.port == Y$	$tcp.port == 80$ (HTTP)
3. $icmp$	shows ping traffic.
4. dns	track domain name resolution.

* Flaregraph

PC1

PC2

nc -l 1337

nc 192.168.0.2 1337

echo "Hello"

Greener starts listening on port 1337

Host A (Client)

Host B (Greener)

192.168.0.1

192.168.0.2

1. [SYN seq=0] →
2. [SYN, ACK] seq=0 ACK=1 ←
3. [ACK] seq=1 ACK=1. →

* Network commands

1. $ifconfig$ / $ipconfig$ → displays net configuration
2. $ping$ → checks connectivity
3. $tracert$ → displays path
4. $arp -a$ → shows mapping btw IP & MAC
5. $netstat$ → shows active network
6. $getmac$ → shows MAC address

* File sharing

1. SSH (Secure Shell) :- secure network protocol used for remote login & secure communication betw 2 comp.
2. scp (secure copy protocol) :- command like cp that securely transfer files betw. comp. using SSH protocol.

Steps :-

1. At Receiver (check SSH service)
⇒ $sudo systemctl status ssh$
↳ checks SSH service running.

⇒ $sudo systemctl start ssh$
if NOT then,

2. scp hello.txt pft@192.168.0.10:8/home/pft.

file to sent → username → destination folder

Enter :- Then asks for password.

Enter Password then successfully file copied.

* Wireless LAN

- Enables to connect without physical cables by using radio frequency (RF) signals.
- Provides mobility, flexibility, ease of deployment, makes suitable for home office, etc.
- follows IEEE 802.11 family of standards, known as Wi-Fi.
- devices communicate through access point (AP) or router.
- uses 802.11 PC service set identifier to identify n/w & security mechanisms WPA2-Psk & WPA-PSK to protect



network from unauthorized access

* Sample configuration

Parameter	Value
SSID	CN-lab
Wireless Mode	802.11n
Security	WPA2-PSK
Pass	root123
Router IP	192.168.0.101
DHCP Range	192.168.0.100 - 192.168.0.200.

Result:-

A local area network was successfully designed & configured. Network connectivity was tested. N/w traffic generated during communication & file transmission was captured & analyzed using Wireshark packet analyzer.

Conclusion:-

The experiment provided practical understanding of LAN design, IP addressing, network connectivity, packet capture, & network traffic analysis. While Wireshark provided detailed packet-level info.

The successful file transmission demonstrated the practical use of LAN for communication & resource sharing.